

C Programming Detailed Notes (Beginner Friendly)

1. Introduction to C Language

C is a procedural and structured programming language developed by Dennis Ritchie in 1972. It is considered the foundation for many modern programming languages such as C++, Java, and Python.

C programs are compiled, meaning the entire code is translated into machine code before execution, making it faster than interpreted languages like Python.

Basic structure of a C program:

```
#include <stdio.h>
int main() {
    printf('Hello World');
    return 0;
}
```

2. Features of C

- Simple and efficient
- Portable (can run on different systems)
- Fast execution (compiled)
- Modular programming (using functions)
- Rich library support

3. Structure of a C Program

1■■ Header files (like stdio.h)

2■■ Global declarations

3■■ main() function

4■■ Variable declarations

5■■ Executable statements

4. Data Types in C

Basic types: int, char, float, double.

Derived types: arrays, structures, pointers, unions.

Void type: represents no value.

Type	Size	Example
int	4 bytes	int age = 18;
float	4 bytes	float pi = 3.14;
double	8 bytes	double rate = 5.5;
char	1 byte	char grade = 'A';

5. Variables, Constants, and Keywords

- Variable: container for storing data.

Example: int a = 5;

- Constant: fixed value that cannot be changed.

Example: const float PI = 3.14;

- Keywords: reserved words like int, float, if, else, for, return.

6. Operators in C

Arithmetic Operators: +, -, *, /, %

Relational Operators: ==, !=, >, <, >=, <=

Logical Operators: &&, ||, !

Assignment Operators: =, +=, -=, *=, /=

Increment/Decrement: ++, --

7. Input and Output Functions

- `printf()` → for output
- `scanf()` → for input

Example:

```
int a;  
scanf('%d', &a);  
printf('%d', a);
```

8. Conditional Statements

****if-else statement:****

```
if(a > b)  
    printf('A is greater');  
else  
    printf('B is greater');
```

****switch statement:****

```
switch(choice){  
case 1: printf('Yes'); break;  
default: printf('No');}
```

9. Loops in C

****for loop:**** used when number of iterations is known.

```
for(int i=1; i<=5; i++) printf('%d', i);
```

****while loop:**** used when condition is checked first.

```
while(i<=5){printf('%d',i); i++;}
```

****do-while loop:**** runs at least once.

```
do{printf('%d',i); i++;}while(i<=5);
```

10. Arrays

Array is a collection of elements of the same type stored in continuous memory.

```
int marks[5] = {90,80,85,70,75};
```

marks[0] → first element (90).

11. Strings

String is a collection of characters ending with '\0'.

```
char name[10] = 'Vishesh';
```

Common string functions (in string.h): strlen(), strcpy(), strcat(), strcmp().

12. Functions

Functions divide code into reusable parts.

Example:

```
int add(int a, int b){return a+b;}
```

```
int main(){printf('%d',add(5,3));}
```

Types: Library functions (like printf) and User-defined functions.

13. Recursion

Recursion is when a function calls itself.

Example:

```
int fact(int n){if(n==0)return 1;else return n*fact(n-1);}
```

14. Pointers

Pointer stores address of another variable.

```
int a=10; int *p=&a; printf('%d',*p);
```

& → address operator, * → value at address (dereferencing).

15. Structures and Unions

Structures combine different data types.

```
struct Student {int roll; char name[20]; float marks;};
```

Unions share memory among all members — only one member can be used at a time.

16. Storage Classes

Types:

- auto (default)
- static (retains value)
- extern (global variable)
- register (stored in CPU)

17. File Handling

Used to read/write files.

```
FILE *fptr;
```

```
fptr=fopen('data.txt','w');
```

```
fprintf(fptr,'Hello');
```

```
fclose(fptr);
```

Modes: 'r' read, 'w' write, 'a' append, 'r+' read/write.

18. Dynamic Memory Allocation

Functions: malloc(), calloc(), realloc(), free() from stdlib.h.

```
int *p = (int*)malloc(5*sizeof(int));
```

19. Preprocessor Directives

Commands that begin with #. Example: **#define PI 3.14**

Also include header files using **#include**.

20. Common Escape Sequences

\n → new line

\t → tab space

\a → alert sound

\b → backspace

\r → carriage return

21. Example Program

```
#include <stdio.h>
int main(){
    int a,b,sum;
    printf('Enter two numbers: ');
    scanf('%d %d', &a, &b);
    sum=a+b;
    printf('Sum = %d', sum);
    return 0;
}
```